

We select an arbitrary positive number ε . For ε we want to find another positive number δ such that the value of the function at any point x from the punctured δ -neighbourhood of point a must be inside the interval $[b-\varepsilon, b+\varepsilon]$ (speaking about any point x we imply only the points x in the domain of the function). If there is such δ for any $\varepsilon > 0$, b is said to be the limit of the function at point a . Otherwise b is not the limit of the function at point a .

Reader: And what does your "otherwise" mean in practice?

Author: Assume that the search for δ has been successful for n diminishing numbers $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$. But then you notice that for a certain number ε' it is impossible to find the required number δ , i.e. for any value of δ no matter how small there is always at least one point x from the punctured δ -neighbourhood of point a at which the value of the function lies outside the interval $[b-\varepsilon', b+\varepsilon']$. Imp. Remember

Reader: But can it happen that we reduce the δ -neighbourhood of point a so much that not a single point x , belonging to the domain of the function, remains in the δ -neighbourhood?

Author: Obviously this is impossible. Because the function is defined over an interval, and point a is taken either from this interval or coincides with its end point.

Reader: Everything seems clear. Apparently, in order to root all this firmly in my mind we should discuss the graph of a function.

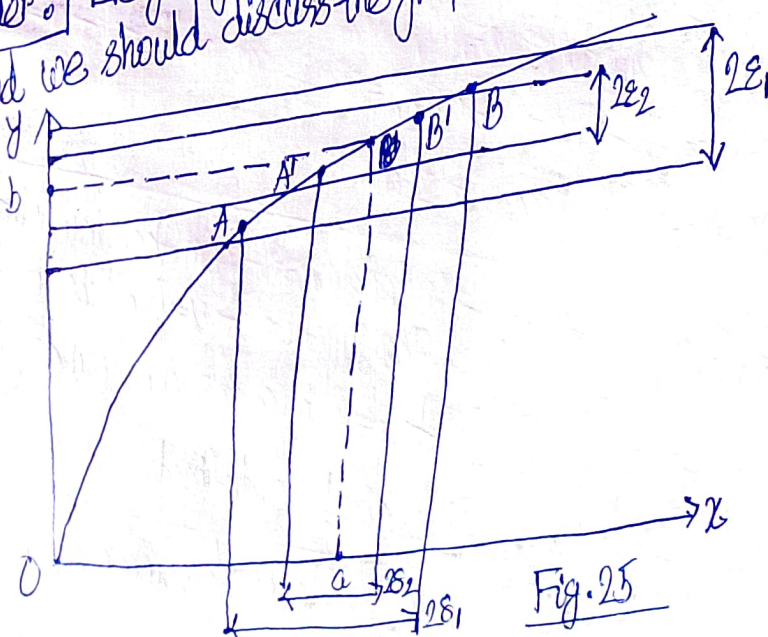


Fig. 25